

Cisco Network Project

Configuration Notes

Platform: Cisco Packet Tracer

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This project demonstrates the design and configuration of a small-business network using Cisco Packet Tracer.

The network uses VLANs to separate different types of devices and services. It includes IPv4 addressing, DHCP, trunking, router-on-a-stick, inter-VLAN routing, basic SSH remote management, DNS, and an internal HTTP website.

1. Assigned Devices

Table 1: Network Devices

Device	Role
R1 — Cisco 1941	Router and inter-VLAN gateway
SW1 — Cisco 2960-24TT	Layer 2 switch
SERVER0	Web and DNS server
PC-IT	IT workstation
PC-EMPLOYEE	Employee workstation
PC-GUEST	Guest workstation
PC-MANAGEMENT	Management workstation

Table 2: VLAN and IP Addressing Plan

VLAN	Name	Network	Gateway
10	IT	10.0.10.0/24	10.0.10.1
20	Employee	10.0.20.0/24	10.0.20.1
30	Server	10.0.30.0/24	10.0.30.1
40	Guest	10.0.40.0/24	10.0.40.1
50	Management	10.0.50.0/24	10.0.50.1

Table 3: Router Interfaces

Interface	IP Address	VLAN
G0/0.10	10.0.10.1	10
G0/0.20	10.0.20.1	20
G0/0.30	10.0.30.1	30
G0/0.40	10.0.40.1	40
G0/0.50	10.0.50.1	50

Table 4: Switch Port Assignments

Switch Port	VLAN	Connected Device
Fa0/1	10	PC-IT
Fa0/2	20	PC-Employee
Fa0/3	40	PC-Guest
Fa0/4	50	PC-Management
Fa0/10	30	Server0
Fa0/24	Trunk	R1 G0/0

Note: Fa0/24 is configured as a trunk between SW1 and R1

2. Router-on-a-stick

R1 uses router-on-a-stick configuration to provide communication between the different VLANs. The physical interface G0/0 connects to SW1 through the trunk on Fa0/24.

Table 5: R1 uses the following subinterfaces

G0/0.10 → VLAN 10 → 10.0.10.1
G0/0.20 → VLAN 20 → 10.0.20.1
G0/0.30 → VLAN 30 → 10.0.30.1
G0/0.40 → VLAN 40 → 10.0.40.1
G0/0.50 → VLAN 50 → 10.0.50.1

Note: Each subinterface uses 802.1Q VLAN tagging.

3. DHCP Configuration

R1 provides DHCP services for the client VLANs. The first 20 addresses in each DHCP subnet were excluded from the DHCP pools. This leaves addresses beginning at .21 available for client devices.

Table 6: DHCP Pools

IT

- Network: 10.0.10.0/24
- Gateway: 10.0.10.1
- DNS: 10.0.30.10

Employee

- Network: 10.0.20.0/24
- Gateway: 10.0.20.1
- DNS: 10.0.30.10

Guest

- Network: 10.0.40.0/24
- Gateway: 10.0.40.1
- DNS: 10.0.30.10

Management

- Network: 10.0.50.0/24
- Gateway: 10.0.50.1
- DNS: 10.0.30.10

4. SERVER0 Configuration

SERVER0 is in VLAN 30. The server uses a static IP address for network services to remain available at a known address.

IP Address:	10.0.30.10	Enabled Services
Subnet Mask:	255.255.255.0	HTTP: Enabled
Default Gateway:	10.0.30.1	DNS: Enabled
DNS Server:	10.0.30.10	

5. Internal Website

An internal website was configured on SERVER0 using the Packet Tracer HTTP service. The website can be accessed using `http://www.wylie.com`.

The DNS server contains an A record that maps:

`www.wylie.com` → 10.0.30.10

www.wylie.com is being used as a local domain inside the Packet Tracer network. It is not a real website.

6. SSH Remote Management

SSH was configured on SW1 and R1 for remote management. A local administrator account was created for authentication.

The VTY lines were configured using

```
line vty 0 15
login local
transport input ssh
```

Purpose of the Commands

`login local` tells the device to authenticate users using the locally configured username and password. `transport input ssh` restricts remote VTY connections to SSH.

This prevents Telnet from being used through the VTY lines.

SSH Test

From PC-MANAGEMENT, SW1 was accessed using: `ssh -l admin 10.0.50.2`

7. Network Verification

- show ip interface brief — confirmed router subinterfaces were up/up.
- show vlan brief — confirmed VLANs and switch access ports.
- show ip dhcp binding — confirmed DHCP leases were assigned.
- Ping tests — confirmed connectivity between VLANs.
- SSH test — confirmed remote management through SSH.
- DNS test — confirmed www.wylie.com resolved to 10.0.30.10.
- HTTP test — confirmed the internal website loaded successfully.

8. Project Conclusion

The completed network represents a small-business environment with separate VLANs for IT, employees, servers, guests, and management.

R1 provides routing between the VLANs and DHCP services for client devices. SW1 provides VLAN segmentation and uses VLAN 50 for management. SERVER0 provides DNS and HTTP services, allowing users to access the internal website through www.wylie.com.

SSH provides remote management access to the Cisco devices while Telnet access is disabled.

The network was tested using Cisco IOS verification commands, ping, DHCP verification, SSH, DNS, and HTTP.